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2 Sources of Uncertainty:

- aggregate
- individual

Aggregate Uncertainty:

$$Y = \bar{z} F(K, N)$$

Idiosyncratic Uncertainty:

- labor income: $w \cdot \varepsilon$

↳ both follow Markov Process

Household Problem:

$$v(k, \varepsilon; z, \Gamma) = \max_{c, k'} \{u(c) + \beta E[v(k', \varepsilon'; z, \Gamma)]\}$$

s.t.

$$c + k' = (1 + r(K, N, z))k + w(K, N, z) \varepsilon$$

$\Gamma' = H(\Gamma, z, z')$ law of motion of Γ

Firm Optimization:

$$w = z(1 - \alpha) K^\alpha N^{-\alpha}$$

$$r = z \alpha K^{\alpha-1} N^{1-\alpha} - \delta$$

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Market Clear:

$$K = \int k d\Gamma(k, \varepsilon)$$

$$N = \int \varepsilon d\Gamma(k, \varepsilon)$$

s.t.

$\Gamma(k, \varepsilon)$ is dist of people over states

Distribution is non-stationary

recursive CE for this economy is

- a set of prices $\{w, r\}$
- households' decision rules + value function
- law of motion for $\Gamma(k, \varepsilon)$, $H(\cdot)$
- set of aggregate variables

s.t.

- 1) given price + $H(\cdot)$, $V(\cdot)$ and policy functions solve HH optimization problem
- 2) policy func's + Markov Processes for ε + z imply that today's Γ is mapped into tomorrow's Γ' by $H(\cdot)$
- 3) aggregate variables satisfy Mcc's
- 4) prices satisfy firm FOCs

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Computational Shortcut - Krusell + Smith (1998):

- $Y = z' K^a N^{1-a}$, $z = \begin{cases} z^b & \text{if bad} \\ z^g & \text{if good} \end{cases}$ w/ transition $\Pi_{z'z}$

- idiosyncratic uncertainty:

$\varepsilon = \begin{cases} 1 & \text{if employed} \\ 0 & \text{otherwise} \end{cases}$ w/ transition $\Pi_{\varepsilon'\varepsilon}$

$z = z^g: \log K' = a_0 + a, \log K$

$z = z^b: \log K' = b_0 + b, \log K$

$$N' = NP(\varepsilon' = 1 \mid \varepsilon = 1) + (1-N)P(\varepsilon' = 1 \mid \varepsilon = 0)$$

new HH problem:

$$v(k, \varepsilon; z, K) = \max_{c, k'} \{u(c) + \beta \sum_{z'} \Pi_{z'z} \sum_{\varepsilon'} \Pi_{\varepsilon'\varepsilon} E[v(k', \varepsilon'; z', K')]\}$$

s.t.

$$c + k' = (1 + r(K, N, z))k + w(K, N, z)\varepsilon$$

$$\log(K') = a_0 + a, \log K \quad \text{if } z = z^g$$

$$\log(K') = b_0 + b, \log K \quad \text{if } z = z^b$$

- guess a_0, a, b_0, b

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Computational Algorithm:

- 1: guess a_0, a_1, b_0, b_1
- 2: solve consumer optimization problem
- 3: using decision rules, compute K and simulate the economy $\Rightarrow$ compute time series for capital
- $\hookrightarrow$ estimate $\hat{a}_0, \hat{a}_1, \hat{b}_0, \hat{b}_1 \Rightarrow$ update a_i, b_i
- 4: go back to (2) and repeat until convergence

know good fit by R^2 ; 0.999 in paper

Kruger, Mitman, Perri (2016):

$\hookrightarrow$ "Macroeconomics + Household Heterogeneity"

- use Krusell-Smith as baseline
- technology: $Y = z F(K, N)$ s.t. $z = \begin{cases} z^l & \text{when recession} \\ z^h & \text{when good} \end{cases}$
- demographic: stochastic aging
 - $\hookrightarrow$ age but only two phases w/ probability to transition from young $\rightarrow$ old
 - $\hookrightarrow$ measure of people is 1

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Kruger, Mitman, Perri, (2016):

- age $j \in \{W, R\}$

$\hookrightarrow W \xrightarrow{1-\theta} R \xrightarrow{1-v} \text{Dead}$

$\hookrightarrow$ each dead agent replaced w/ agent at age W .

- π_W : stationary measure of the young

- π_R : stationary measure of the old

$\hookrightarrow \pi_W + \pi_R = 1$

$$\pi_W = \pi_W \theta + \pi_R (1-v)$$

$$\pi_R = 1 - \pi_W$$

$$\pi_W = \pi_W \theta + (1 - \pi_W)(1-v)$$

$$\pi_W (1 - \theta + 1 - v) = 1 - v$$

$$\pi_W = \frac{1-v}{2-\theta+v} \quad \therefore \pi_R = 1 - \frac{1-v}{2-\theta+v} = \frac{1-\theta}{2-\theta+v}$$

Idiosyncratic Risk:

1) Unemployment shock $s = \{u, e\}$

$$\pi(s'|s, z', z)$$

$\pi_z(s)$ is invariant distribution

2) labor productivity $\pi(y'|y)$

$\pi(y)$ is invariant dist.

$$\sum \pi(y) y = 1$$

3) people differ in β

Government:

- Unemployment Insurance

↳ income of unemployed: $\rho \cdot w(z, \Gamma) \cdot y = b(y, z, \Gamma)$

→ ρ is replacement rate, b is unemp benefits.